

State of the Programme — 17 July 2026

What this is: Bill Gaede's Rope Hypothesis developed into a machine-verified research corpus: 122 registered claims (108 code-backed, all passing), 61 papers, 76 benchmarks. Every claim carries a status — Derived / Modeled / EFT-constrained / Conjecture / Failed / Open — and failures are kept as findings.

Scope (PM-005): a theory of ORDINARY MATTER — the first generation (electron, up, down), their composites, and the photon — plus emergent relativity and a developing gravity sector. It claims nothing about heavier generations.

The derived core, by sector

- **EM / Maxwell:** charge = winding (topological); Maxwell structure from network mechanics; absolute normalization (Σ) measured, not derived.
- **Optics:** 10/10 phenomena derived; linear light exactly blind to extensibility.
- **Chemistry:** bond mechanism quantitative (mode overlap); $\sigma > \pi$; bond length from the core; parameter-free H₂ vibration check passes at 16%; molecular GEOMETRY opened (CHEM-GEO-001): heteronuclear additivity derived (blind: 7–16% high, contraction = next order); phase-blocking + orthogonality theorems give bent H₂O (derived; H₂S at 92° supports); CO₂ collinear (Modeled).
- **Nuclear:** Yukawa FORM exact (screened-mode Green's function); atomic masses predicted C-12–U-238 within 0.15% of mass with ONE calibrated constant; residuals 80–86% traced to declared quantum omissions.
- **Magnetism:** E/M as one rope (along/around); force signs derived — BOTH halves: magnetic (gap geometry, EM-012) and electrostatic (constraint-source superposition, EM-015); permanent magnets. Field-tensor dictionary assembled (EM-016); GG-005 upgraded Conjecture → Modeled.
- **Matter:** atoms sit at the impenetrability coverage threshold (fixes N); the mesh scale a is proven irreducible.
- **Relativity:** emergent, transverse-sector only; fundamental preferred frame; matter cannot access the lattice frame.
- **Gravity:** MATURE NEGATIVE RESULT — Newtonian mechanism derived (Poisson/ $1/r$ from force balance; flux-conservation $1/r^2$); zero-point theorem (uniform vacuum is flat); classical Einstein completion FALSIFIED within the defined response class ($\gamma \in [-1, 0]$ vs measured 1; 0.44" vs 1.75"; no-monopole lemma; sourcing exhaustion under explicit hypotheses H1–H6); quantum-induced completion the leading conjecture, underived.
- **Particle masses:** structure derived (massless photon; composition rule; mass = mode energy); values are inputs — proven, not presumed (knot-count refuted for leptons; excitation spectra fail without tuning).
- **Quantum boundary:** local accounts theorem-blocked ($\text{CHSH} \leq 2$, derived); the substrate has 4 of 5 necessary conditions for a nonlocal (pilot-wave-type) account — the fifth (Born-equilibrium statistics) and all sufficient conditions absent.

The constants ledger

Derived: forms, ratios, signs, thresholds, composition rules, one geometric surface coefficient ($a_S/a_V = 1.108$ vs 1.130), $\gamma = 1/\sin^2\theta$. **Inputs (irreducible or open):** mesh scale a ; $\hbar$; electron mass; nucleon mass unit; k 's value; Σ (bounded by experiment); G (provably reduces to the mass question); the nuclear bond depth ε (calibrated once).

Experiment: the live account

- **PVLAS excluded our identification** ($\Sigma \approx 1.5 \times 10^{25}$ overshoot the birefringence bound $\sim 570\times$): the corrected bound is $\Sigma \gtrsim 8.6 \times 10^{27} (k/T_0 - 1) \text{ J/m}^3$; ATLAS light-by-light is genuine QED. *The theory has taken a real loss against real data.*
- **Live prediction:** vacuum birefringence with NEGATIVE index shifts in a 3:1 ratio (QED: positive, 7:4) — sitting in the window between current bounds and QED sensitivity. Decided by experiment, not preference.
- **The vacuum-tension problem:** Σ implies an effective vacuum density $\geq \sim 10^{11} \text{ kg/m}^3$ that must not gravitate. Static half resolved by the zero-point theorem (holds for any Σ); dynamic/cosmological half open.

The honest walls

Gravity tensor structure (the next campaign); Born rule + quantum equilibrium (two miracles; approach only with a pre-committed bar); $\hbar$ (do not chase); hierarchy = the mass question (one problem, not two); cosmological zero point; K_c derivation.

Corrections on the record (credibility assets)

Twist-gap mechanism retracted within one turn (gauge argument forbids it); Coulomb-instantaneity bonus self-downgraded; QB-006 corrected within one turn (Mark: the statistical fifth condition); PVLAS correction of the Σ identification; three stale Σ references caught and superseded in this consolidation pass.

Method

Human physical judgment (scope calls, postulate decisions, independent-determination demands, the Born-rule correction) + machine verification (compute before prose; anti-fitting bars fixed in advance; negatives registered). The loop is the asset.